EAST WEST UNIVERSITY

Project Report

Title: Design a full-fledged network for an organization with multiple subnets.

Course Title: Computer Networks

Course Code: CSE405

Section: 02

SUBMITTED BY:

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Introduction

Apex University, is an enterprise like East West University, owns many computers, with a complex network infrastructure. Apart from wired internet access to all the classrooms, labs, employee PCs, library and other administrative and academic wings, the university also provides wireless internet access for every campus. On top of that the university runs complex networked systems to support several of its business process like admissions, advising, results, eTender, library management, accounts and so on.

This complex network infrastructure is subnetted and switching/routing mechanisms are in practice.

Tools

1. CISCO Packet Tracer 8.2.2

2. PC–PT, Laptop-PT, SmartPhone-PT

3. Router (Generic Router-PT)

4. Wireless (AccessPoint-PT)

5. Switch (2960 IO S 15)

6. Server (DNS Server, DHCP Server, Web Server)

7. Connector (Copper Straight-Through, Serial DCE)

Tasks

1. The web page of the university reflecting its own information.

2. A single DNS sever is installed to locate web server - meaning people will browse University's web site from any campus with the following address: http://www.apex.edu.bd

3. Configured the whole network in such a way that IP for the hosts of different campuses is automatically assigned by a single DHCP server.

4. Created five LAN network, one for each campus; among the hosts in a network making sure some wireless hosts are added in addition to wired hosts.

5. University's full network has covered its five campuses with five routers; Connections between the campus routers are established strictly following the given topology.

6. Connectivity between all the hosts are established.

7. For routing protocol, dynamic routing algorithm i.e., OSPF is used.

8. IP addresses are assigned as instructed: Campus-1, Campus-2, and Campus-8 Network addresses are from the “A” class; Campus-3, Campus-4, and Campus-5 Network addresses are from “C” class, and Campus-6 and Campus-7 are from “B” class. Placement of all 3 servers i.e., WEB server, DNS server, and DHCP server are in campus-5.

9. The Connections between all the routers are in serial connections.

10. Further expansion and growth are possible if need be.

Network Diagram

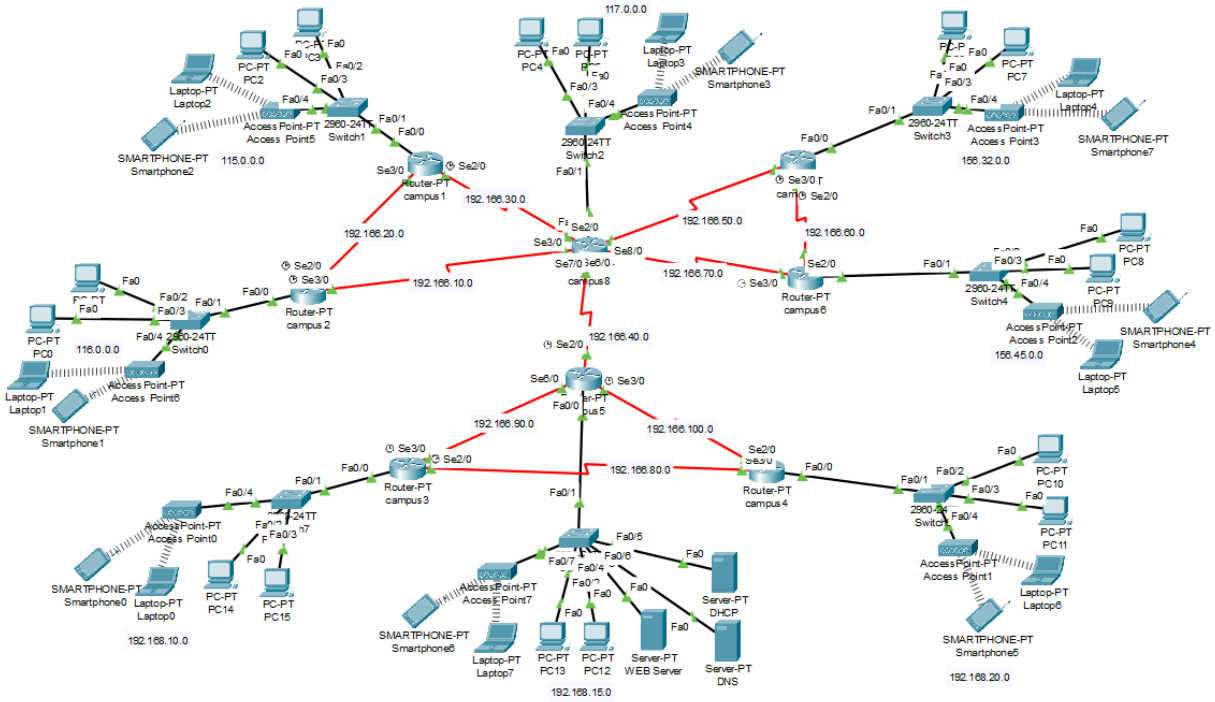

Figure-1

Router Setup

Router1-8 are for Network1-8 in respective order.

Networks 1, 2, and 8 are assigned to class A network where IP 115.0.0.0 is for network 1, IP 116.0.0.0 is for network 2, and IP 117.0.0.0 is for network 8.

Networks 6 and 7 are assigned to class B networks where IP 156.45.0.0 is for network 6, and IP 156.32.0.0 is for network 7.

Networks 3, 4, and 5 are assigned to class C networks where the network IP is 192.168.10.0 is for network 3, IP 192.168.15.0 is for network 5, and IP 192.168.20.0 is for network 4.

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Routers and intra-router networks are set up exactly as shown in Figure-1.

CLI commands used to set each router are given below:

```
#Campus 1
interface fa0/0
ip address 115.0.0.254 255.0.0.0
no shut
do wr
exit

interface se3/0
ip address 192.166.20.2 255.255.255.0
no shut
do wr
exit

interface se2/0
ip address 192.166.30.1 255.255.255.0
clock rate 64000
no shut
```

```
do wr
exit

#Campus 2
interface fa0/0
ip address 116.0.0.254 255.0.0.0
no shut
do wr
exit

interface se2/0
ip address 192.166.20.1 255.255.255.0
clock rate 64000
no shut
do wr
exit

interface se3/0
ip address 192.166.10.1 255.255.255.0
clock rate 64000
no shut
do wr
exit

#Campus 3
interface fa0/0
ip address 192.168.10.254 255.255.255.0
no shut
do wr
exit
```

```
interface se2/0
ip address 192.166.80.1 255.255.255.0
clock rate 64000
no shut
do wr
exit

interface se3/0
ip address 192.166.90.2 255.255.255.0
clock rate 64000
no shut
do wr
exit

#Campus 4
interface fa0/0
ip address 192.168.20.254 255.255.255.0
no shut
do wr
exit

interface se2/0
ip address 192.166.100.1 255.255.255.0
no shut
do wr
exit

interface se3/0
ip address 192.166.80.2 255.255.255.0
```

```
no shut
do wr
exit

#Campus 5
interface fa0/0
ip address 192.168.15.254 255.255.255.0
no shut
do wr
exit

interface se2/0
ip address 192.166.40.2 255.255.255.0
clock rate 64000
no shut
do wr
exit

interface se3/0
ip address 192.166.100.2 255.255.255.0
clock rate 64000
no shut
do wr
exit

interface se6/0
ip address 192.166.90.1 255.255.255.0
no shut
do wr
exit
```

```
#Campus 6
interface fa0/0
ip address 156.45.0.254 255.255.0.0
no shut
do wr
exit

interface se2/0
ip address 192.166.60.1 255.255.255.0
no shut
do wr
exit

interface se3/0
ip address 192.166.70.2 255.255.255.0
clock rate 64000
no shut
do wr
exit

#Campus 7
interface fa0/0
ip address 156.32.0.254 255.255.0.0
no shut
do wr
exit

interface se2/0
ip address 192.166.60.2 255.255.255.0
```

```
clock rate 64000
no shut
do wr
exit

interface se3/0
ip address 192.166.50.2 255.255.255.0
clock rate 64000
no shut
do wr
exit

#Campus 8
interface fa0/0
ip address 117.0.0.254 255.0.0.0
no shut
do wr
exit

interface se2/0
ip address 192.166.30.2 255.255.255.0
no shut
do wr
exit

interface se3/0
ip address 192.166.10.2 255.255.255.0
no shut
do wr
exit
```

```
interface se7/0
ip address 192.166.40.1 255.255.255.0
no shut
do wr
exit

interface se6/0
ip address 192.166.70.1 255.255.255.0
no shut
do wr
exit

interface se8/0
ip address 192.166.50.1 255.255.255.0
no shut
do wr
exit
```

Routing tables were setup using OSPG dynamic routing algorithm.

```
#Campus 1
router ospf 1
network 115.0.0.0 0.255.255.255 area 1
network 192.166.20.0 0.0.0.255 area 1
network 192.166.30.0 0.0.0.255 area 1
exit
```

```
#Campus 2
router ospf 2
network 116.0.0.0 0.255.255.255 area 1
network 192.166.20.0 0.0.0.255 area 1
network 192.166.10.0 0.0.0.255 area 1
exit

#Campus 3
router ospf 3
network 192.168.10.0 0.0.0.255 area 1
network 192.166.90.0 0.0.0.255 area 1
network 192.166.80.0 0.0.0.255 area 1
exit

#Campus 4
router ospf 4
network 192.168.20.0 0.0.0.255 area 1
network 192.166.100.0 0.0.0.255 area 1
network 192.166.80.0 0.0.0.255 area 1
exit

#Campus 5
router ospf 5
network 192.166.40.0 0.0.0.255 area 1
network 192.166.100.0 0.0.0.255 area 1
network 192.166.90.0 0.0.0.255 area 1
network 192.168.15.0 0.0.0.255 area 1
```

```
exit
#Campus 6
router ospf 6
network 192.166.60.0 0.0.0.255 area 1
network 192.166.70.0 0.0.0.255 area 1
network 156.45.0.0 0.0.255.255 area 1
exit


#Campus 7
router ospf 7
network 192.166.60.0 0.0.0.255 area 1
network 192.166.50.0 0.0.0.255 area 1
network 156.32.0.0 0.0.255.255 area 1
exit

#Campus 8
router ospf 8
network 192.166.10.0 0.0.0.255 area 1
network 192.166.30.0 0.0.0.255 area 1
network 192.166.70.0 0.0.0.255 area 1
network 192.166.50.0 0.0.0.255 area 1
network 117.0.0.0 0.255.255.255 area 1
network 192.166.40.0 0.0.0.255 area 1
exit
```

DHCP server is located at network 5 and is tasked with automatically assigning IP addresses to hosts. A similar IP helper is commanded onto every network.

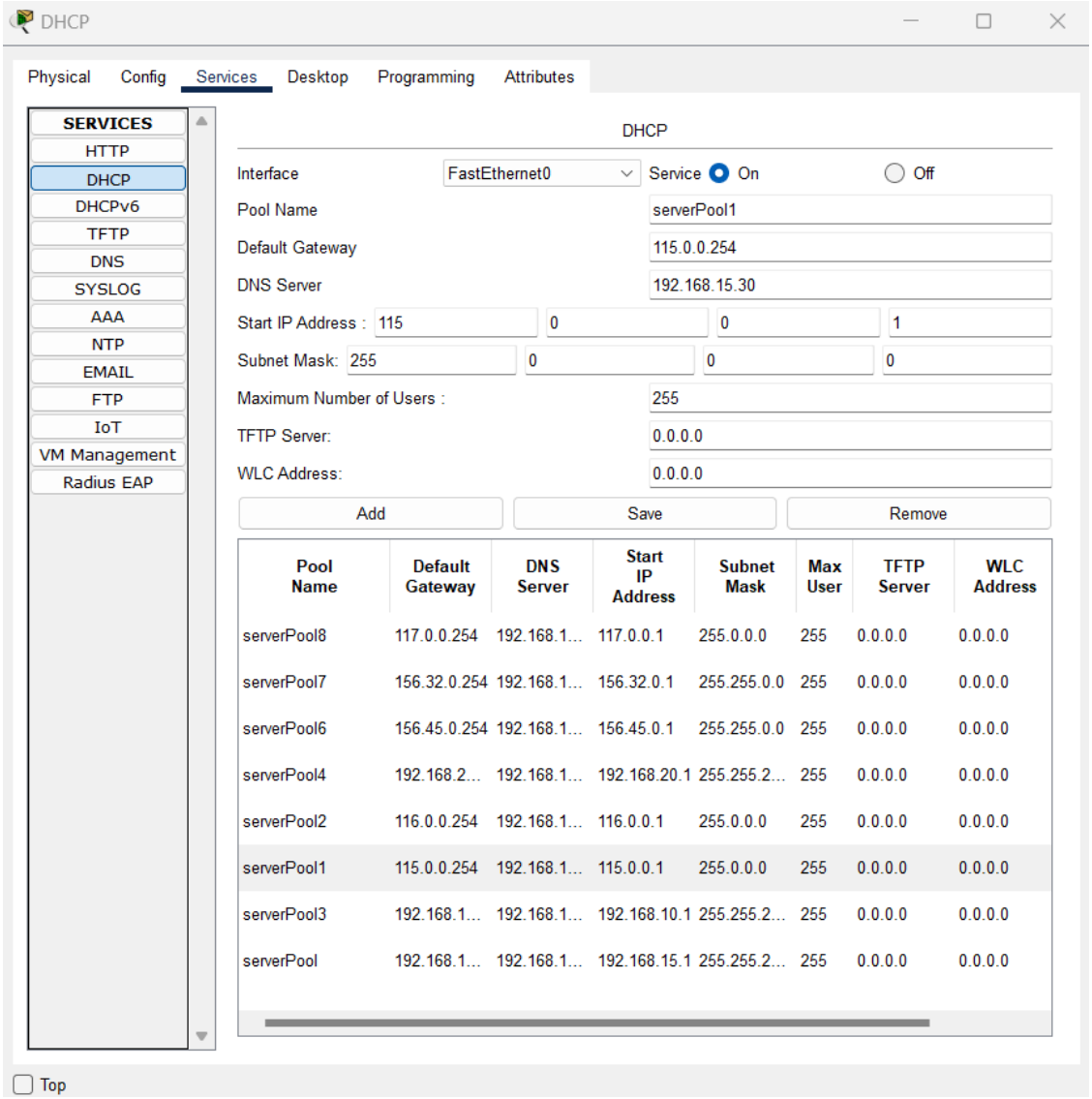
Command:

interface fa0/0

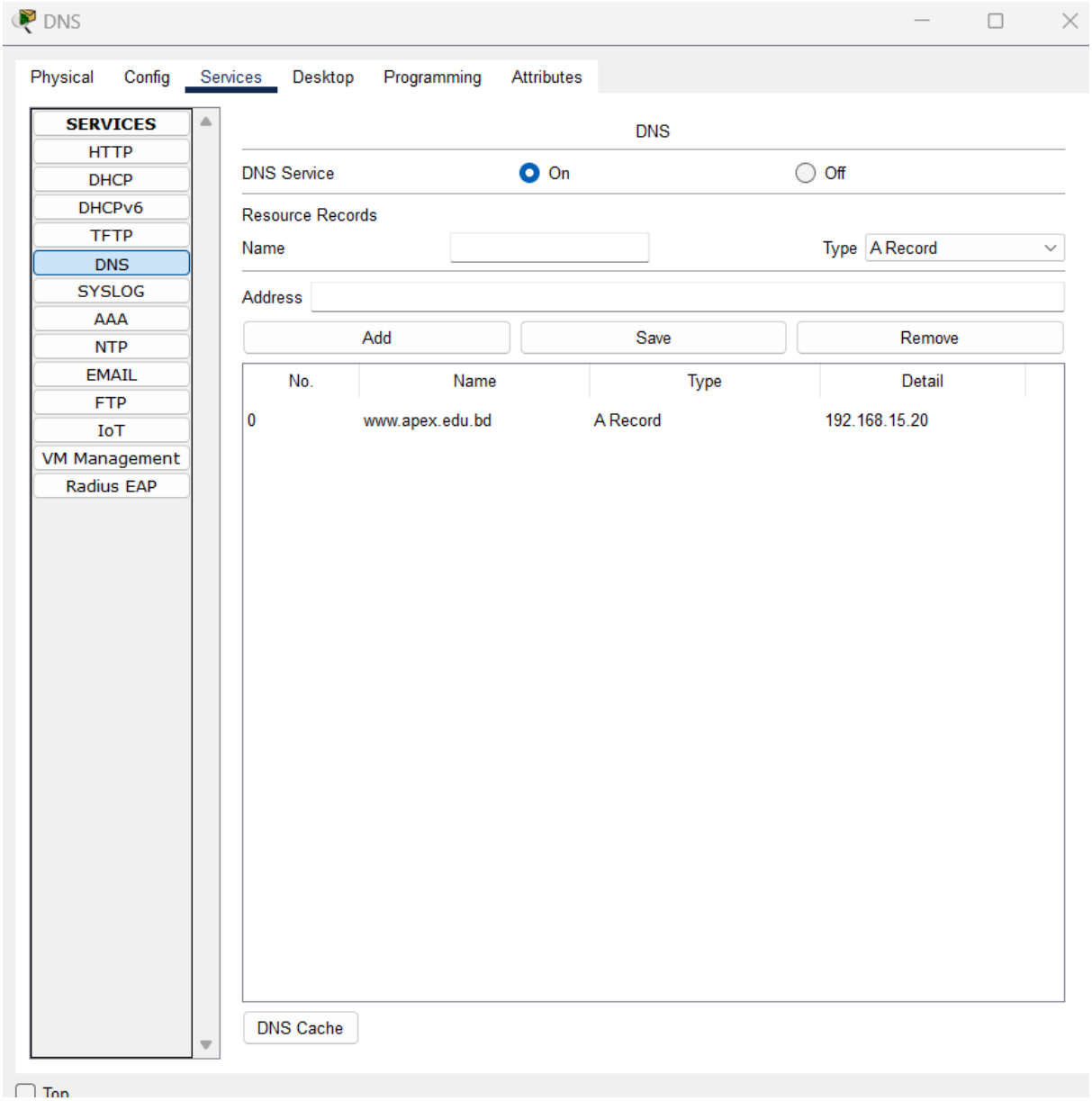
ip helper-address 192.168.15.25

exit

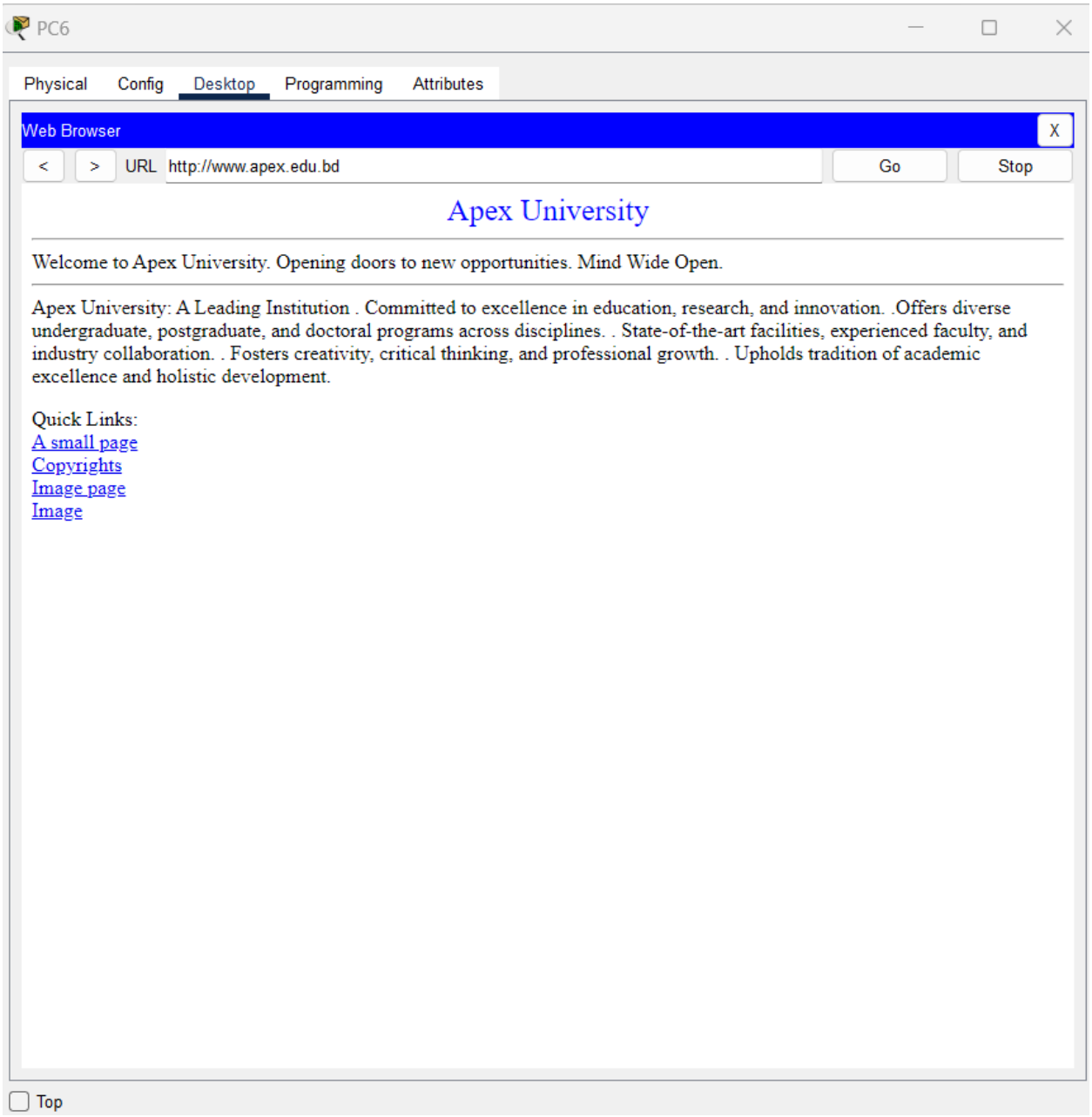
DHCP Server Setup



DNS Server Setup



Web Page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Conclusion

In conclusion, university network systems play a crucial role in the functioning and operation of modern educational institutions. Class schedules, online course materials, and email are just a few of the vital information and resources that these systems may give staff, instructors, and students. Along with facilitating administrative duties like grading and record-keeping, they also allow members of the university community to communicate and work together. All things considered, the effectiveness and success of any educational institution depend on a carefully planned and managed university network infrastructure.